Cheran Rajamanikam

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Professional Summary

Artificial Intelligence and Machine Learning undergraduate at Visvesvaraya Technological University (First Class Distinction) with hands-on experience building Generative AI, LLM, and Retrieval-Augmented Generation (RAG) solutions in Python. Co-author of a peer-reviewed IEEE paper on AI-assisted biosensor analysis. Experienced in machine learning model development, data analytics, REST APIs, and AWS cloud architecture. Seeking an AI Research Intern or Associate Engineer Intern role at IBM.

Technical Skills

- **Programming Languages**: Python, JavaScript, SQL
- **Machine Learning and AI**: Machine Learning, Deep Learning, Natural Language Processing, Computer Vision, TensorFlow, Keras, scikit-learn, Hugging Face Transformers, LangChain
- **Generative AI**: Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), Prompt Engineering, FAISS, Vector Databases
- **Cloud and Backend**: AWS (Solutions Architecture), Django, REST APIs
- **Data and Analytics**: Pandas, NumPy, Matplotlib, Jupyter, Data Analysis, Feature Engineering
- **Tools**: Git, GitHub, VS Code, Unreal Engine

Education

Bachelor of Engineering, Artificial Intelligence and Machine Learning Visvesvaraya Technological University, Bengaluru | Sep 2024 - Sep 2028 (expected)

- First Class Distinction
- Relevant coursework: Machine Learning, Deep Learning, Computer Vision, Data Structures and Algorithms, Probability and Statistics

Pre-University Course (Physics, Chemistry, Biology, Mathematics) Christ University, Bengaluru | Jun 2022 - May 2024

- Grade: 82%

Experience

Artificial Intelligence Intern Decodelabs | Remote | 2025 - Present

- Built and deployed production-ready Generative AI pipelines using LLMs and prompt optimization, improving response relevance through systematic evaluation across benchmarks
- Implemented data analytics workflows to extract insights from large unstructured datasets, supporting data-driven product decisions
- Collaborated with cross-functional engineering teams to integrate AI modules into existing software, shortening feature delivery cycles
- Prototyped state-of-the-art GenAI tools and frameworks and wrote internal technical documentation adopted by the team

Research and Publications

Alpha-Synuclein Biosensing with Dual-Mode Plasmonic Micro-Ring Biosensor for Early-Stage Parkinson's Disease Diagnosis IEEE | Nov 2025 | Co-author

- Co-authored a peer-reviewed paper proposing a dual-mode plasmonic micro-ring biosensor for ultra-sensitive detection of alpha-synuclein biomarkers linked to Parkinson's disease
- Applied signal processing and sensor data analysis to validate biosensor performance and demonstrate potential for early diagnosis using AI-assisted spectral analysis

Projects

ResearchMind AI | LLM Application

- Developed an LLM-based research assistant that lets users explore research-related information through natural-language interaction
- Integrated large language model capabilities into an accessible application to support AI-assisted research workflows

AI Study Planner | Full-Stack Web Application

- Developed a full-stack study planning application with a functional frontend and backend to help students organize study activities and manage tasks
- Integrated frontend and backend components to deliver a working learning management experience with structured study schedules

Moto Companion | Team Project

- Contributed to an end-to-end motorcycle safety and assistance application, focusing on AI/ML and data processing
- Application scope covers emergency SOS, crash detection, ride matching, and community engagement features

Intelligent RAG-Powered Knowledge Assistant | 2025

- Architected a Retrieval-Augmented Generation system over domain-specific document corpora, using embedding-based semantic search with FAISS vector stores for contextual question answering
- Integrated multilingual transformer models to support regional-language queries, demonstrating cross-lingual NLP

Certifications

- AI Fundamentals: Foundations for Understanding AI, IBM SkillsBuild, May 2026
- Lifelong Professional Skills, IBM SkillsBuild, May 2026
- AWS Solutions Architecture Job Simulation, Amazon Web Services via Forage, Dec 2024

Languages

English (Native/Bilingual), Hindi (Full Professional), Tamil (Native), Kannada(Native/Bilingual).